TEAM U-17

SIGNIFICANT

Sign Language App for Social Inclusivity & Well-Being

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BACKGROUND

Widespread lack of familiarity with **Sign Language**

1% of population rely on SL, remaining **99%** do not understand SL

Multilingualism & **Various** Sign Languages

contributes to **Communication Obstacles** between deaf & hearing communities

Social Isolation & **Emotional Distress**

could cause suicidal depression, substance abuse or violent behaviour

SOLUTION

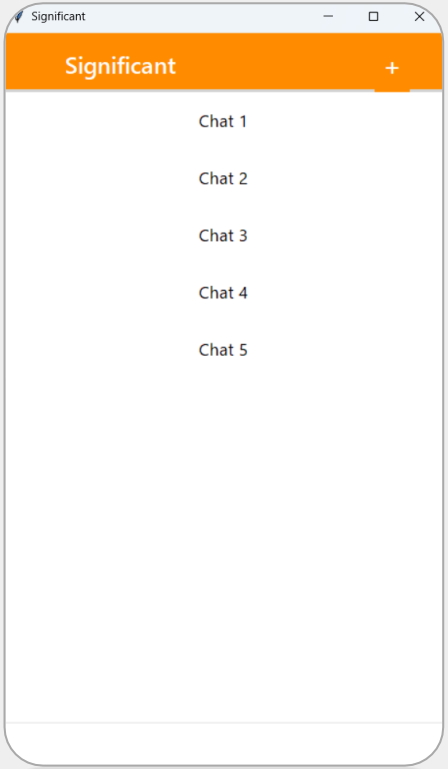
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Multiple Language Translation (01)

Sign Language to Spoken Language (02)

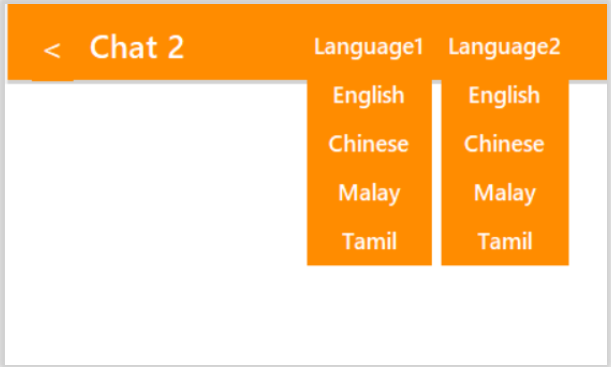
(03) Speech-to-Text

(04) Text-to-Speech



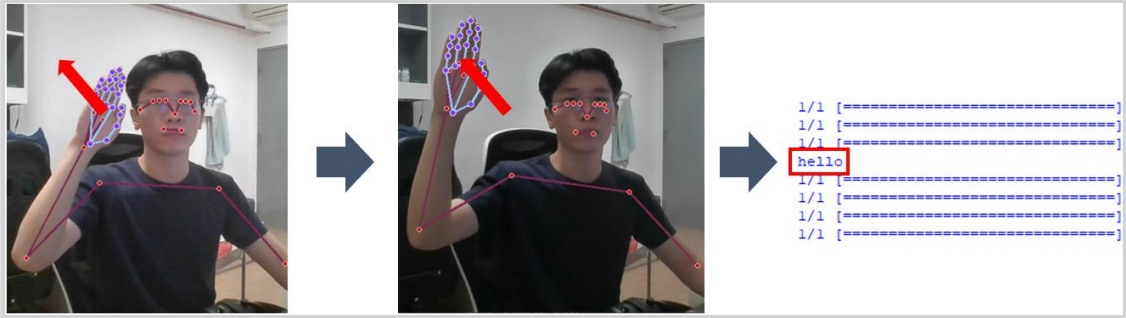
MAIN FUNCTIONS

MULTIPLE LANGUAGE TRANSLATION



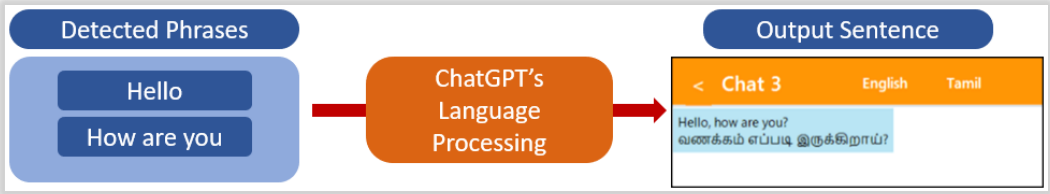
- Select Language 1 & 2
- Both languages will be displayed
- English, Chinese, Malay, and Tamil are included

SIGN ⇨ SPOKEN LANGUAGE



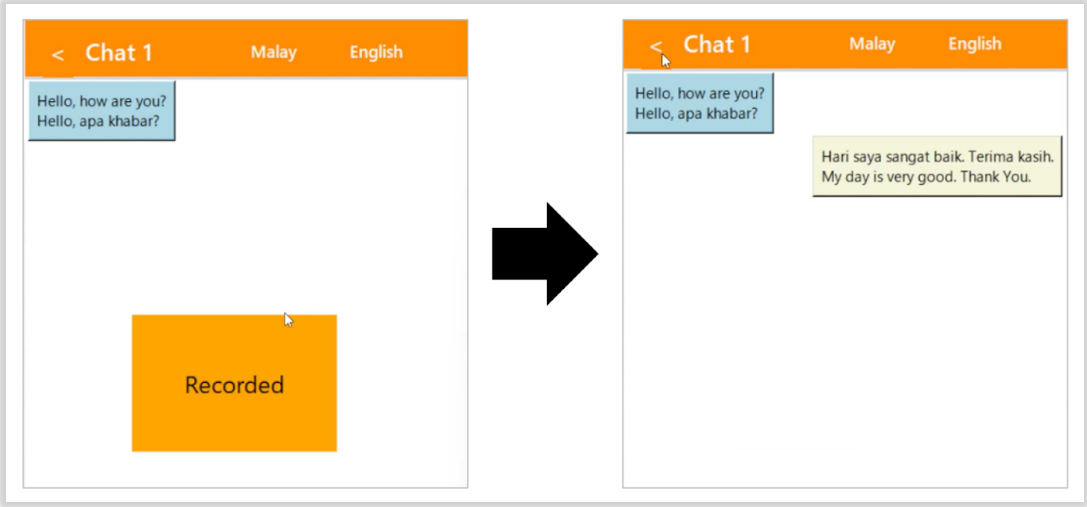
- ASL ⇨ Text
- Motion Tracking
- Action Detection

- Additional refinement ⇨ **ChatGPT's** processing capabilities



- Complete Sentences
- Accurate Punctuation & Grammar

SPEECH-to-TEXT



- Spoken Language ⇨ Text
- Facilitates communication
- Provide typing option



TEXT-to-SPEECH



- **Google's Text to Speech API**
- Chat bubbles can be Voiced Out
- Crucial for those who have difficulties reading the text

MATERIAL



Python



MediaPipe



OpenCV



Keras



Google API
Speech-to-Text



Google API
Text-to-Speech

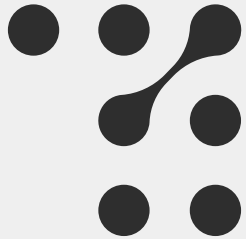


Google
Translate API



ChatGPT API

ULTIMATE GOAL



SL becomes as accessible as other spoken languages

like how Google Translate works for spoken languages

Communication Barriers & Social Isolation

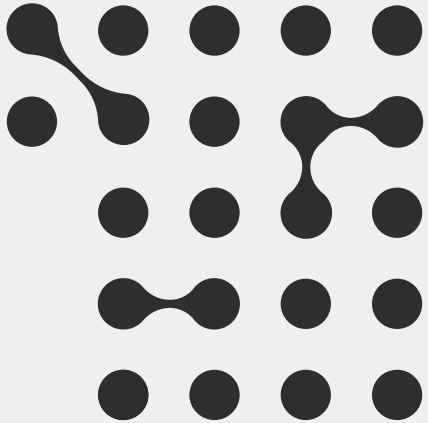
can be significantly reduced or even eliminated

Transformative Step for Humanity

in fostering Resilient Cities that embrace diversity and inclusivity

THANKS

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Questions & Answers

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